



OPEN Explainable AI for lung cancer detection via a custom CNN on CT images

Mohamed Hammad^{1,2}✉, Mohammed ElAffendi¹, Ahmed A. Abd El-Latif^{1,3}, Abdelhamied A. Ateya^{1,4}, Gauhar Ali¹ & Pawel Plawiak^{5,6}✉

Lung cancer, which claims 1.8 million lives annually, is still one of the leading causes of cancer-related deaths globally. Patients with lung cancer frequently have a bad prognosis because of late-stage detection, which severely limits treatment options and decreases survival rates. Early detection is essential for better outcomes, but traditional CT image analysis is time-consuming, prone to error, and relies on subjective judgments. To overcome these issues, we propose a custom convolutional neural network (CNN) combined with explainable AI (XAI) techniques, particularly gradient-weighted class activation mapping (Grad-CAM). This approach is intended to reliably classify lung cancer into squamous cell carcinoma, large cell carcinoma, or adenocarcinoma. Unlike conventional methods, our approach not only achieves highly accurate classification of lung cancer subtypes but also incorporates clinically validated interpretability features to ensure alignment with medical diagnostics. Our model trained on a comprehensive dataset of CT images achieved an overall accuracy of 93.06%. This performance demonstrates the model's robustness in detecting even subtle malignancies, with strong precision, recall, and F1-scores across all cancer types. Including interpretable Grad-CAM visualizations ensures reliability and transparency, aiding clinicians in understanding the model's predictions. This innovative method demonstrates the potential to revolutionize early lung cancer detection and improve patient survival rates by combining state-of-the-art accuracy with explainability tailored for clinical application.

Keywords Lung cancer detection, CNN, Explainable AI, CT images, Grad-CAM, Medical imaging

Lung cancer ranks among the most common and deadly forms of the disease, contributing significantly to global cancer-related deaths¹. According to the World Health Organization, it remains the primary cause of cancer-related deaths, with approximately 2.21 million new cases and 1.8 million deaths each year². The high death toll is due primarily to diagnoses made at advanced stages, where treatment options are limited and less effective³. Detecting lung cancer early is essential, as it greatly increases the likelihood of successful treatment and increases survival rates⁴. This work is driven by the goal of improving early lung cancer detection through advanced computational techniques, with the aim of lowering mortality rates and enhancing patient outcomes. This study specifically addresses the critical need for early, accurate, and interpretable diagnostic systems, which remains a major gap in existing methods.

There are various forms of lung cancer, each with distinct histological and clinical characteristics⁵. Large cell carcinoma, adenocarcinoma, and squamous cell carcinoma are the *three* primary forms⁶. Large cell carcinoma is a less common kind of non-small cell lung cancer (NSCLC) characterized by large, abnormal-looking cells that tend to grow and spread quickly⁷. Nearly 40% of all occurrences of lung cancer are adenocarcinomas, making them the most common kind of disease. It originates in mucus-secreting glands in the lungs and often affects nonsmokers and younger individuals^{8,9}. Squamous cell carcinoma arises from the flat cells lining the airways

¹ELIAS Data Science Lab, College of Computer and Information Sciences, Center of Excellence in Quantum and Intelligent Computing, Prince Sultan University, Riyadh 11586, Saudi Arabia. ²Department of Information Technology, Faculty of Computers and Information, Menoufia University, Shibin El Kom 32511, Egypt. ³Department of Mathematics and Computer Science, Faculty of Science, Menoufia University, Shibin El Kom 32511, Egypt. ⁴Department of Electronics and Communications Engineering, Zagazig University, Zagazig, Egypt. ⁵Department of Computer Science, Faculty of Computer Science and Telecommunications, Cracow University of Technology, Warszawska 24 Str., Kraków 31-155, Poland. ⁶Institute of Theoretical and Applied Informatics, Polish Academy of Sciences, Bałtycka 5, Gliwice 44-100, Poland. ✉email: mhammad@psu.edu.sa; mohammed.adel@ci.menofia.edu.eg; plawiak.pawel@gmail.com

and is strongly associated with smoking. This type of cancer typically grows near the central parts of the lungs and may form cavities¹⁰.

Traditional methods for categorizing lung cancer from CT images have relied primarily on visual inspection by radiologists¹¹. This manual process involves examining the morphological features of nodules, such as size, shape, and density, to determine their degree of malignancy. However, these traditional methods have several disadvantages. First, these methods are time-consuming and require significant expertise, which can lead to interobserver variability and potential misdiagnosis. Moreover, subtle differences in early-stage tumors are often challenging to discern, leading to missed detection. Computer-aided diagnosis (CAD) systems have been developed to address these limitations, offering automated and more consistent analysis of medical images^{11,12}. While these advancements are promising, current CAD systems still fall short in effectively identifying early-stage malignancies and providing actionable insights for clinicians.

The emergence of artificial intelligence (AI), especially deep learning (DL), has transformed CAD systems, increasing their accuracy and reliability¹³. Traditional AI approaches, including rule-based systems and conventional machine learning algorithms, have struggled with the complexities and high dimensionality of medical image data^{14–20}. For instance, recent studies have demonstrated the utility of DL in diverse medical imaging applications, such as mammogram synthesis for breast cancer diagnosis¹⁷, enhancing synthetic mammogram image quality for improved detection¹⁸, accurate skin cancer diagnosis through tailored architectures such as Smart MobiNet¹⁹, and optimizing detection using ensemble approaches²⁰.

On the other hand, the extraction of hierarchical features from images has shown to be a highly effective use of DL approaches, particularly convolutional neural networks (CNNs)^{21–23}. CNNs are exceptionally well suited for medical imaging because of their ability to learn spatial hierarchies of features, which enables them to detect subtle patterns indicative of different types of lung cancer^{24–26}. The application of CNNs has resulted in significant improvements in both the accuracy and efficiency of lung cancer detection from CT scans, surpassing the capabilities of traditional AI methods^{27,28}.

Despite the progress made with deep learning-based CAD systems, recent studies in the literature have revealed several challenges and limitations^{29–39}. Many of these studies have reported suboptimal diagnostic accuracy, especially when dealing with small or ambiguous nodules. Furthermore, the therapeutic adoption of deep learning models is frequently hampered by their “black-box” nature, which makes it difficult for doctors to accept and interpret the model’s judgments in the absence of knowing the underlying reasoning. These challenges include the lack of explainability, limited generalizability across diverse patient populations, and the massive datasets required for training. This lack of explainability limits the clinical utility of these systems, as healthcare professionals require transparent decision-making processes to ensure patient safety and effective communication.

This study directly addresses these gaps by proposing a novel lung cancer detection approach that combines a tailored CNN architecture with explainable AI (XAI), specifically gradient-weighted class activation mapping (Grad-CAM). This work advances the state of the art in AI-driven CAD systems by bridging the gap between high diagnostic accuracy and model interpretability. The incorporation of XAI allows clinicians to visualize and understand the CT images that most influence the model’s predictions, thereby enhancing trust in the system. The explainability provided by Grad-CAM ensures that clinicians can trust and effectively utilize the model in real-world practice. Our key contributions include the following:

- Our method focuses on detecting subtle changes in lung tissue indicative of early-stage malignancies, which are often challenging for existing methods. By identifying these early markers, our approach facilitates timely interventions, enabling less invasive treatments and significantly improving patient survival rates.
- To overcome the “black-box” limitation, our approach incorporates Grad-CAM, enabling clinicians to visualize the regions in CT images that most influence the model’s predictions. This transparency fosters trust in the model’s decisions and facilitates clinician-patient communication, ultimately improving treatment adherence and outcomes.
- Unlike conventional CNNs, our architecture is tailored for the specific complexities of lung cancer detection. It incorporates advanced regularization techniques and optimized feature extraction pipelines, addressing limitations in prior methods and achieving superior diagnostic accuracy.
- The proposed system is designed to generalize across diverse patient populations by leveraging a robust training process and extensive data augmentation strategies. This ensures consistent and reliable performance in real-world clinical settings.

Literature review

Recently, deep learning techniques, especially CNNs, have become increasingly prominent in medical imaging, with a particular focus on lung cancer detection^{29–39}. The literature on this topic can be broadly categorized into studies addressing binary classification problems^{29–33}, where the focus is on distinguishing between cancerous and noncancerous cases, and those addressing multiclass classification^{34–39}, which involves differentiating between various types of lung cancer. Each category of study presents unique challenges and opportunities, with ongoing research striving to increase the accuracy, robustness, and interpretability of the models.

Binary classification methods for lung cancer detection

Binary classification methods aim to distinguish between malignant and benign lung nodules via CT images. Several studies have employed CNNs to achieve this task, leveraging their ability to automatically extract relevant features from medical images. For example, Gopinath et al.²⁹ used a deep CNN (DCNN) and saliency maps for lung CT scan analysis, replacing the standard neural network with a cat-optimized CNN. The model’s was 99.89% using the LIDC-IDRI dataset for binary classification. A multimodal fusion deep neural network

(MFDNN), created by S.K.B. et al.³⁰, integrates genetics, clinical data, and medical imaging to enhance lung cancer diagnosis with a 92.5% accuracy. Asuntha and Srinivasan³¹ aimed to detect and classify lung cancer severity via DL methods. They employed advanced feature extraction techniques such as HoG, wavelet transforms, and LBP, followed by fuzzy particle swarm optimization (FPSO) for feature selection. A unique FPSOCNN is used to classify the features, which decreases computing complexity and outperforms previous methods. Shafi et al.³² introduced a DL-enabled SVM-based cancer diagnostic model for detecting lung cancer in CT scans. The model achieves 94% accuracy in early-stage pulmonary nodule detection, outperforming existing methods. Jena et al.³³ presented a method to enhance early disease diagnosis and accuracy using the LIDC dataset. It involves noise reduction with Gaussian and Wiener filters, region-growing segmentation, and feature extraction. A deep Gaussian mixture model in a region-based CNN (DGMM-RBCNN) is used for dimensionality reduction, achieving 87.79% accuracy. However, these approaches often face significant challenges. One major limitation is the imbalanced nature of medical datasets, where benign cases typically outnumber malignant cases, leading to biased model training. Furthermore, binary classification lacks comprehensive information about the type of lung cancer, which is critical for developing effective treatment regimens. Moreover, the lack of interpretability in these models often results in clinicians being hesitant to rely solely on AI predictions for critical decision-making.

Multiclass classification methods for lung cancer detection

Multiclass classification methods extend beyond binary classification by identifying specific subtypes of lung cancer, such as large cell carcinoma, adenocarcinoma, and squamous cell carcinoma. This classification is essential for personalized treatment planning, as different types of lung cancer may respond differently to treatment. Recent studies, such as those by Lanjewar et al.³⁴, Islam et al.³⁵, Komputer et al.³⁶, Alsheikhy et al.³⁷ and others^{38,39}, have explored the use of deep CNN architectures for this purpose. Lanjewar et al.³⁴ used the Kaggle chest CT-scan dataset to classify lung cancer into four categories via a modified DenseNet201 model. By adding layers and applying feature selection methods, the model was tested with various ML classifiers. It achieved an average accuracy of 95% with a p-value below 0.001, highlighting the effectiveness of machine learning in improving lung cancer diagnosis from CT scans. Islam et al.³⁵ introduced the DCNN-GRU model, which combines a deep CNN (DCNN) with gated recurrent units (GRUs) for better detection and classification of lung abnormalities. This model incorporates explainable AI techniques, including LIME, SHAP, and Grad-CAM, to enhance interpretability. Testing on COVID-19 and lung cancer datasets showed high accuracy rates of 99.30% and 98.97%, respectively. Komputer et al.³⁶ utilized a dataset of 1,000 lung CT-scan images across four categories to evaluate various transfer learning models. EfficientNetB3 outperformed the other methods, achieving 97.78% accuracy, 97.34% precision, 98.33% recall, and a 97.78% F1 score, demonstrating its superior effectiveness in lung cancer classification. This study highlights the potential of EfficientNetB3 for early diagnosis and its ability to improve diagnostic processes in lung cancer management. Alsheikhy et al.³⁷ presented a fully automated system combining VGG-19 and LSTM networks for early lung cancer detection and classification, achieving over 98.8% accuracy. When tested on the Kaggle and LUNA16 datasets, the model reached an average accuracy of 99.42%, with recall, precision, and F scores exceeding 99.7%. Despite these advancements, multiclass classification models face significant challenges, including the high intraclass variability of cancer subtypes and the scarcity of annotated datasets for certain rare subtypes. Furthermore, the increased complexity of multiclass models often exacerbates the “black-box” problem, making it difficult to interpret the model’s decision-making process. Table 1 summarizes all the previous methods for detecting lung cancer and compares their performance, results, and limitations.

Authors	Methodology	Classes	Results	Limitations
Gopinath et al. ²⁹	Deep CNN (DCNN) with saliency maps	Binary (malignant vs. benign)	99.89% accuracy on LIDC-IDRI dataset	Imbalanced dataset issue, lack of subtype classification and interpretability
S.K.B et al. ³⁰	Multimodal Fusion Deep Neural Network (MFDNN)	Binary (malignant vs. benign)	92.5% accuracy	May struggle with complex feature integration and model interpretability
Asuntha and Srinivasan ³¹	FPSO-CNN with advanced feature extraction	Binary (malignant vs. benign)	99.23% average accuracy	Limited to binary classification, potential lack of interpretability
Shafi et al. ³²	Deep learning-enabled SVM-based model	Binary (malignant vs. benign)	94% accuracy in early-stage nodule detection	May not address subtype classification, limited interpretability
Jena et al. ³³	DGMM-RBCNN (Deep Gaussian Mixture Model with Region-based CNN)	Binary (malignant vs. benign)	87.79% accuracy	Imbalanced dataset, lack of subtype classification, and interpretability issues
Lanjewar et al. ³⁴	Modified DenseNet201 with feature selection	Multiclass (adenocarcinoma, large cell, squamous, normal)	95% average accuracy, p value < 0.001	High intraclass variability, limited dataset diversity
Islam et al. ³⁵	DCNN-GRU with Explainable AI techniques (LIME, SHAP, Grad-CAM)	Multiclass (COVID-19, lung cancer, normal)	99.30% accuracy for COVID-19, 98.97% for lung cancer	Complex model interpretability, high computational requirements
Komputer et al. ³⁶	EfficientNetB3 transfer learning model	Multiclass (adenocarcinoma, large cell, squamous, normal)	97.78% accuracy, 97.34% precision, 98.33% recall, 97.78% F1-Score	Complexity of model, dataset size limitations
Alsheikhy et al. ³⁷	VGG-19 combined with LSTM networks	Multiclass (adenocarcinoma, large cell, squamous, normal)	Over 98.8% accuracy, 99.42% average accuracy, recall, precision, and F score > 99.7%	Complexity of model, high computational demand

Table 1. Comparison of lung Cancer detection methods: performance, results, and limitations.

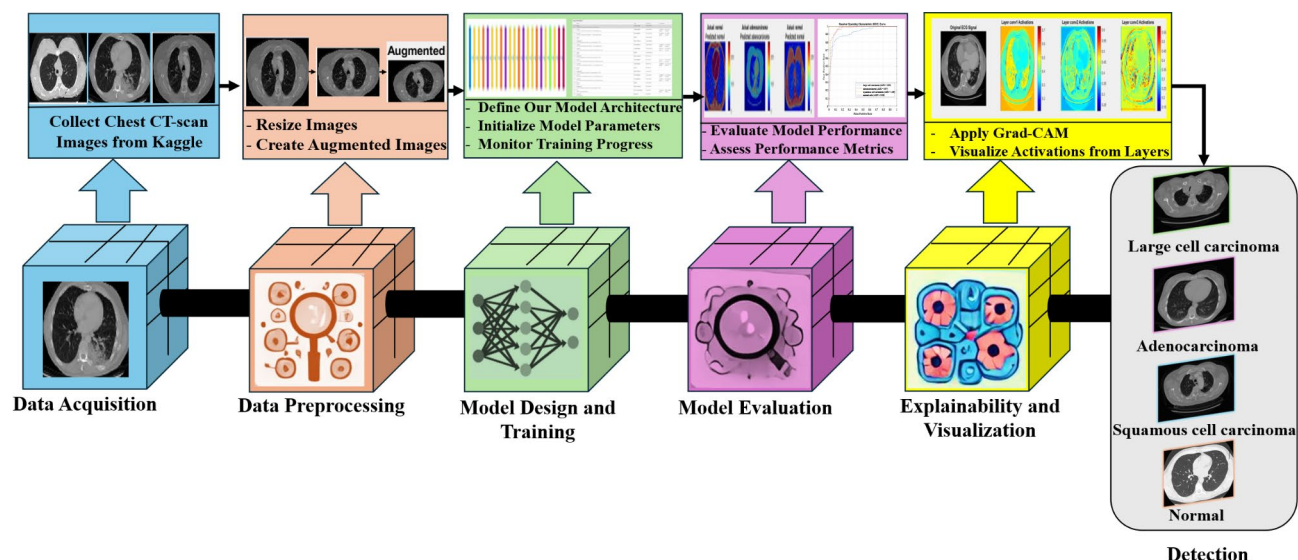


Fig. 1. Block diagram of our method.

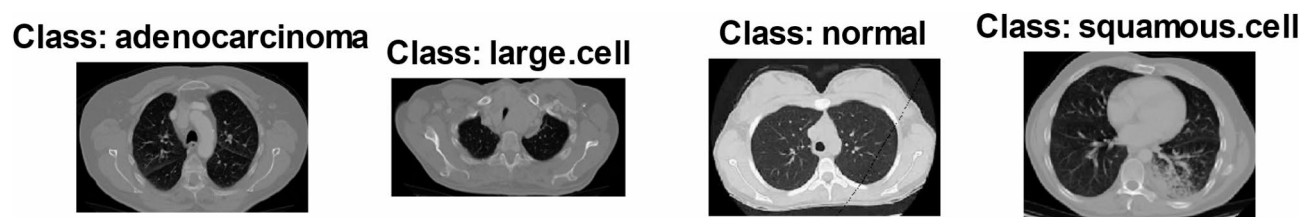


Fig. 2. Sample images from the dataset.

Overcoming limitations with our proposed method

Our proposed method aims to address the limitations identified in existing binary and multiclass classification approaches for lung cancer detection. By designing a custom CNN architecture tailored to the specific characteristics of lung cancer, we seek to improve both classification accuracy and model generalizability across different patient populations. One of the key innovations of our approach is the integration of XAI techniques, particularly Grad-CAM. Grad-CAM offers visual insights into the model's predictions by highlighting the areas of CT images that have the greatest impact on the decision-making process. This not only improves the transparency of our model but also fosters trust among clinicians, thereby supporting the adoption of AI in clinical settings. Additionally, our model addresses the challenge of data imbalance by employing advanced data augmentation techniques to mitigate the impact of class imbalance on model training. By incorporating a comprehensive multiclass classification framework, our method is capable of accurately distinguishing between various lung cancer subtypes, providing valuable diagnostic information that can guide treatment decisions. The use of a custom CNN allows for the fine-tuning of network parameters to optimize performance for specific lung cancer features, thereby overcoming the generalization issues faced by existing models.

Methodology

The methodology of this study outlines a comprehensive framework for the automated detection of lung cancer types via DL. The approach involves several stages, each focusing on distinct aspects of the data processing pipeline, including data preparation, model architecture design, training, evaluation, and explainability. This section details the procedures and techniques employed, ensuring a robust and transparent model development process. The overall process is depicted in the general block diagram shown in Fig. 1. This block diagram illustrates the sequential steps from data acquisition through model evaluation.

Data acquisition and preparation

The dataset utilized in this study comprises chest CT scan images sourced from Kaggle⁴⁰. The images are formatted in JPG or PNG and are categorized into *four* main classes: squamous cell carcinoma, large cell carcinoma, adenocarcinoma and normal cells. The dataset is organized into *three* primary directories—training, validation, and testing—each containing subdirectories for different cancer types and normal cells. The *imageDatastore* function is employed to load the images, ensuring the preservation of the folder structure for labeling. Sample images from each class's dataset are displayed in Fig. 2.

An augmentation strategy is implemented on the training set via the *imageDataAugmenter* function. This strategy incorporates random rotations (± 10 degrees), translations (± 10 pixels along the X and Y axes), and scaling (90–110%), thereby introducing variations that the model may encounter in real-world scenarios. These augmentation parameters were carefully selected to simulate realistic variability while preserving the anatomical integrity of the CT scans. Random rotations of ± 10 degrees address slight orientation changes due to patient positioning or scanner alignment. Translations of ± 10 pixels along the X and Y axes account for small shifts in image positioning during scanning or preprocessing. Scaling within the range of 90–110% reflects minor changes in image magnification that may arise from scanner settings or differences in patient size. The augmentation strategy enhances the model's robustness and generalization ability, enabling it to perform effectively on diverse imaging conditions. To ensure uniformity across all samples, the augmented data are resized to a consistent input size of 256×256 pixels. This resizing step eliminates inconsistencies in image dimensions, enabling efficient processing by CNN. Additionally, the images are converted to grayscale, which simplifies the computational process by reducing the input dimensionality from *three* colour channels (RGB) to a single intensity channel. This grayscale conversion emphasizes texture and structural patterns critical features for distinguishing between different types of cancerous and normal tissues without compromising diagnostic accuracy. Figure 3 illustrates the augmentation process, showing the variations introduced to a sample image.

Model architecture

The custom CNN architecture employed in our study introduces several enhancements over conventional designs to address the unique challenges of lung cancer detection in CT images. The network begins with an *imageInputLayer* that accepts input images of size 256×256 , with no initial normalization to preserve the original image characteristics. The subsequent layers include the following:

- **Convolutional Layers:** These layers apply multiple filters to the input images, extracting essential features. Each convolutional operation is followed by another to enhance the feature extraction capability, particularly for detecting subtle malignancies. The configuration includes 3×3 filters, with an increasing number of filters across layers (8, 16, 32, 64, and 128) to progressively extract more complex features. The choice of 3×3 filters is based on their ability to capture fine-grained details while maintaining computational efficiency. The increasing number of filters allows the network to learn hierarchical features, from simple edges and textures to more complex patterns indicative of different types of lung cancer.
- **Batch normalization and activation:** After each convolutional operation, a *batch normalization layer* is included to standardize the activations and improve training stability. Next, nonlinearity is added using a rectified linear unit (ReLU) layer, which enables the network to learn increasingly intricate patterns. The use of ReLU is justified by its ability to mitigate the vanishing gradient problem and accelerate convergence.
- **Pooling Layers:** To reduce the spatial dimensions and computational load, *MaxPooling2DLayer* layers with a stride of 2 are used, which also helps make the detection more robust to positional variations. Max pooling was chosen over other pooling methods (e.g., average pooling) because it preserves the most salient features, which is crucial for detecting subtle malignancies in lung CT images.
- **Dropout Layer:** Our architecture employs a high dropout rate (0.9) during training, which effectively mitigates overfitting and enhances generalizability across diverse datasets. This is crucial given the potential variability in lung cancer imaging datasets.
- **Fully Connected Layers:** A fully connected layer with 256 neurons replaces the last convolutional layers and integrates the collected information. This layer size balances the model complexity and computational efficiency, ensuring that the network can effectively classify the input images without overfitting.
- **Output Layers:** The output layer consists of a fully connected layer equal to the number of classes in the dataset. It is followed by a classification layer that makes final predictions and a SoftMax layer that creates probability distributions. Figure 4 shows the CNN model's architecture, highlighting each layer and its role in feature extraction and classification.

The training process is configured using the *trainingOptions* function with the 'Adam' optimizer, which is chosen for its efficiency in handling large-scale data. The Adam optimizer ensures efficient and stable convergence

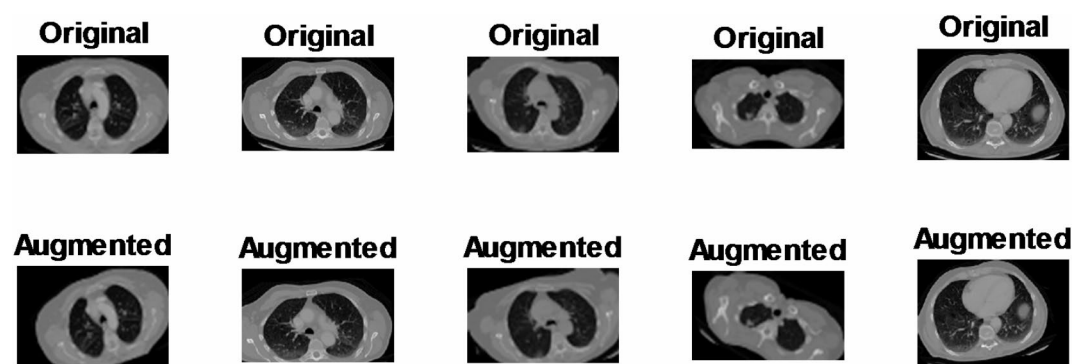


Fig. 3. Effect of augmentation techniques on a sample of images.

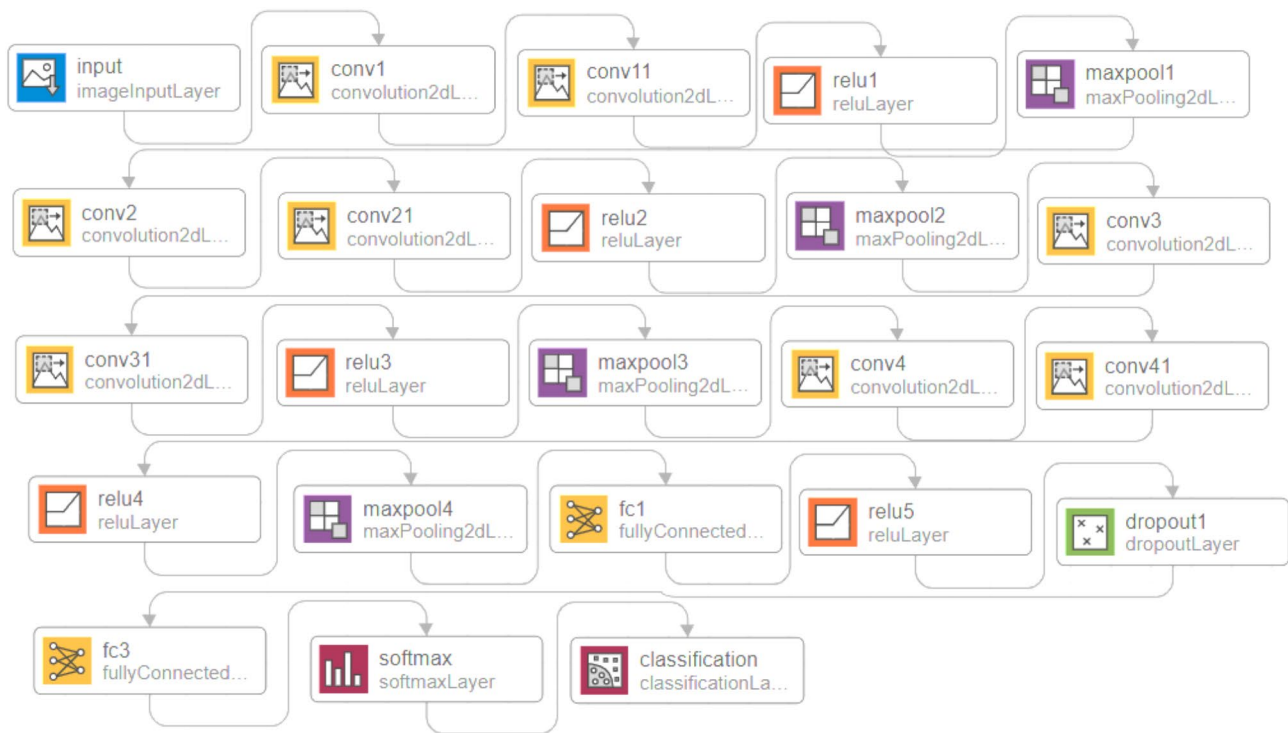


Fig. 4. Architecture of the CNN model.

Hyperparameter	Value
Optimization algorithm	Adam
MiniBatchSize	16
MaxEpochs	80
InitialLearnRate	1e-4
LearnRateSchedule	Piecewise
LearnRateDropFactor	0.5
LearnRateDropPeriod	10
ValidationFrequency	87
Verbose	True
Shuffle	Every-epoch

Table 2. Summary of the key hyperparameters used for training our model.

by dynamically adjusting the learning rate during training, making it suitable for complex medical imaging datasets. The settings include:

- Mini-Batch Size: Set to 16 to ensure manageable memory usage and effective gradient updates. A size of 16 allows for manageable memory usage while maintaining effective gradient updates, ensuring a balance between performance and hardware limitations.
- Max Epochs: Set to 80, allowing sufficient iterations for model convergence.
- Initial Learning Rate: Set at 1e-4, with a piecewise schedule for learning rate adjustments, decreasing by a factor of 0.5 every 10 epochs. This approach ensures effective initial learning while fine-tuning the weights during later stages of training.
- Validation: A validation dataset is utilized to track the model's performance throughout training, with the validation frequency set to 87 iterations. Regular validation every 87 iterations allows for early identification of overfitting or underperformance, ensuring a robust training process. The hyperparameters for the proposed custom CNN architecture were carefully selected to address the unique challenges of lung cancer detection in CT images, including the need for interpretability, high accuracy, and computational efficiency. These choices were validated through iterative experimentation and are summarized in Table 2. Post training, the model is evaluated on the test set.

Optimized hyperparameters, including a mini-batch size of 16 and a piecewise learning rate schedule, are specifically tailored to improve model convergence for medical image analysis.

Explainability and visualization

In complex DL models, particularly in medical imaging, ensuring transparency and interpretability of the model's predictions is crucial for clinical acceptance and trust⁴¹. We employed an explainability technique, especially Grad-CAM, to enhance the interpretability of the predictions made by our DL model. This approach helps pinpoint areas where the model can be enhanced and offers insights into the decision-making process of the model⁴².

In explainable AI, Grad-CAM is a popular technique that generates visual explanations for judgments made by CNNs^{42,43}. Grad-CAM uses the gradients of a target concept, such as the expected class, flowing into the last convolutional layer to build a localization map that highlights the significant regions in the input image. In order to accomplish this, a weighted combination of feature maps is computed, with the weights coming from the gradients about the anticipated class. This produces a heatmap overlay showing the areas of the image that have the most impact on the model's output.

In this study, Grad-CAM was applied to the output of the final convolutional layer for each input image. The resulting heatmaps were then superimposed on the original CT scan images, providing a visual representation of the areas that contributed most to the classification of different types of lung cancer or normal cells. This technique proved invaluable in verifying whether the model focused on relevant anatomical structures, such as the presence of nodules or masses, that are characteristic of specific cancer types.

To gain a deeper understanding of the hierarchical feature extraction process, activations from intermediate layers of the CNN were visualized. By examining these activations, we can observe the progression from low-level features, such as edges and textures, to more complex structures relevant to the classification task. For example, initial layers captured basic patterns such as lines and curves, whereas deeper layers encoded more abstract features, such as shapes and textures indicative of cancerous tissues. These visualizations served multiple purposes:

- **Understanding Feature Hierarchies:** They help in understanding how the model constructs a hierarchy of features, gradually moving from simple to complex representations. This insight is critical for assessing the model's ability to generalize and detect subtle differences between various types of lung cancer.
- **Identifying Potential Model Weaknesses:** By scrutinizing these activations, we identified potential weaknesses in the model, such as overfitting to irrelevant features or failing to capture essential characteristics. For example, if certain critical regions in an image are not highlighted, the model might not be able to effectively learn all relevant features, prompting further refinement of the model architecture or training process.
- **Enhancing Trust and Acceptance:** Providing visual explanations for the model's decisions enhances transparency, making it easier for medical professionals to trust and accept the AI system. By demonstrating that the model's focus aligns with clinically relevant regions, we can build confidence in its predictions and potential for clinical deployment.

The integration of these explainability techniques into the diagnostic workflow has significant implications for clinical practice. By elucidating the decision-making process, we can better communicate model predictions to healthcare providers, facilitating a more informed and collaborative decision-making process. Moreover, visual explanations can assist radiologists in validating the model's findings against their expert judgment, leading to a synergistic relationship between AI tools and medical professionals. Algorithm A outlines the process of developing our DL model for lung cancer detection, incorporating data preparation, augmentation, model training, evaluation, and explainability.

Performance metrics

It is crucial to evaluate deep learning DL models for lung cancer detection and classification to ensure their accuracy and reliability. This section outlines the evaluation metrics used to assess the model's ability to detect and classify various types of lung cancer. These metrics include the area under the receiver operating characteristic curve (AUC-ROC), recall, F1 score, specificity, accuracy, and precision.

A. **Accuracy:** This metric provides a general sense of how well the model performs across all classes.

$$Accuracy = \frac{TrP + TrN}{TrP + TrN + FlP + FlN} \quad (1)$$

where TrP represents the number of correctly identified cancer cases, TrN denotes the number of accurately classified normal cases, FlP provides the number of normal cases misclassified as cancer, and FlN is the number of cancer cases misclassified as normal.

B. **Precision (Positive Predictive Value):** Indicates the proportion of predicted positive cases that are actually positive.

$$Precision = \frac{TrP}{TrP + FlP} \quad (2)$$

C. Recall (Sensitivity): This reflects the model's ability to detect positive cases.

$$Recall = \frac{TrP}{TrP + FlN} \quad (3)$$

D. The F1 score, which is the harmonic mean of recall and precision, offers a single statistic that balances the two metrics.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

E. Specificity (True Negative Rate): Indicates how well the model avoids false positives.

$$Specificity = \frac{TrN}{TrN + FlP} \quad (5)$$

F. The AUC-ROC score illustrates the relationship between the true positive rate (Recall) and the false positive rate (1 - Specificity) at different threshold levels.

$$AUC_ROC = \int_0^1 ROC(t) dt \quad (6)$$

where $ROC(t)$ is the true positive rate at threshold t .

```

1: Start
2: Input: Lung cancer image dataset  $D = \{I_1, I_2, \dots, I_n\}$ 
3: Output: Classified lung cancer images with explainability visualizations
4:  $D_{processed} = preprocess(D)$ 
   - Resize images to 256×256 pixels
   - Convert images to grayscale
5:  $D_{augmented} = apply\_augmentation(D_{processed})$ 
   - Randomly apply transformations (rotation, translation, scaling)
6: Initialize Deep Learning Model with architecture:
   - Input Layer: 256×256×1 images
   - Convolutional Layers: Filters = {32, 64, 128, 256}
   - Activation: ReLU
   - MaxPooling after each convolutional layer
   - Fully Connected Layers for classification
7: Compile Model with optimizer Adam( $\eta$ ) and loss function  $L = -\sum y \log(\hat{y})$ 
8: Train the model:
   for epoch = 1 to max_epochs do
       Train_Model (DeepLearningModel, Train_DS)
   end for
9:  $P = DeepLearningModel.predict(Test\_DS)$ 
   - Evaluate accuracy, precision, recall, and F1 score.
10: Generate Grad-CAM visualizations:
   for each test image  $I$  do
       GradCAM $I$  = generate_GradCAM (DeepLearningModel,  $I$ )
   end for
11: Visualize intermediate activations:
   for each layer  $L$  in the model do
       Activations $L$  = extract_activations (DeepLearningModel,  $L$ ,  $I$ )
   end for
12: Refine the model based on explainability results and clinical feedback
13: End

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Algorithm A. Algorithm for lung cancer detection using deep learning with explainability techniques.

Results and analysis

A presentation and analysis of the results from the model that we employed to categorize and identify lung cancer can be found in this section. A machine with a 2.40 GHz Intel Core i5 CPU, 24 GB of RAM, and a 16 GB NVIDIA GeForce GTX 1650 GPU was used to generate the model using MATLAB R2019b. It was trained for 100 epochs with a mini-batch size of 16 and an initial learning rate of 1×10^{-4} . The model showed consistent performance measures as it converged within the prescribed epochs on the training and validation datasets. By modifying the learning rate, the model was able to effectively prevent overfitting and generalize.

The confusion matrix, displayed in Fig. 5, offers a detailed breakdown of our lung cancer detection model's performance on the test set across the four classes. The model achieved perfect classification for adenocarcinoma and normal cells, with 100% of these cases being correctly identified. The specificity for both adenocarcinoma and normal cells was also 100%, demonstrating that the model effectively avoided false positives in these categories. However, the model showed some misclassifications for large and squamous cell carcinoma. Specifically, 3 instances of large cell carcinoma were incorrectly classified as adenocarcinoma, resulting in a sensitivity of 85.7% and a false negative rate of 14.3% for large cell carcinoma. Similarly, 2 cases of squamous cell carcinoma were misclassified as adenocarcinoma, yielding a sensitivity of 86.7% and a false negative rate of 13.3%. Despite these errors, the overall performance is strong, with sensitivity values reflecting high detection rates for each class and an overall accuracy of 93.06%. These results suggest that while the model is highly effective, particularly for adenocarcinoma and normal cells, there is room for improvement in distinguishing between large and squamous cell carcinoma. To further investigate the misclassification cases, we utilized Grad-CAM visualizations to analyse the regions that influenced the model's incorrect predictions for large cell carcinoma and squamous cell carcinoma. Analysis of the three instances of large cell carcinoma misclassified as adenocarcinoma revealed that the heatmaps predominantly highlighted broad and diffuse image regions. This observation suggests that the heterogeneous and expansive characteristics of large cell carcinoma posed challenges for the model in distinguishing these features from the more structured tumor patterns typically associated with adenocarcinoma. Similarly, for the two cases of squamous cell carcinoma misclassified as adenocarcinoma, the heatmaps demonstrated activation in regions with overlapping morphological features. These regions were characterized by irregular cellular patterns and blurred margins, traits that can occasionally be shared between squamous cell carcinoma and adenocarcinoma, thereby contributing to the model's confusion. Furthermore, additional factors such as variability in CT image quality, subtle morphological differences, and an insufficiently diverse training dataset likely exacerbated the misclassification issues.

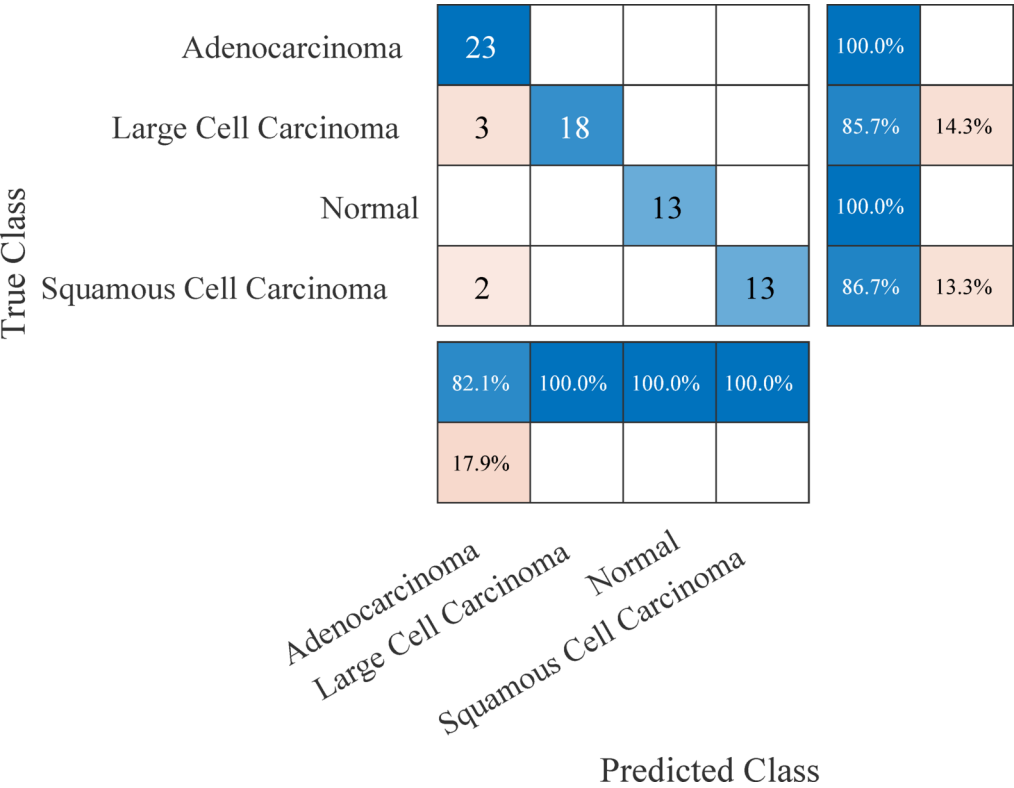


Fig. 5. Confusion matrix of our model on the test set.

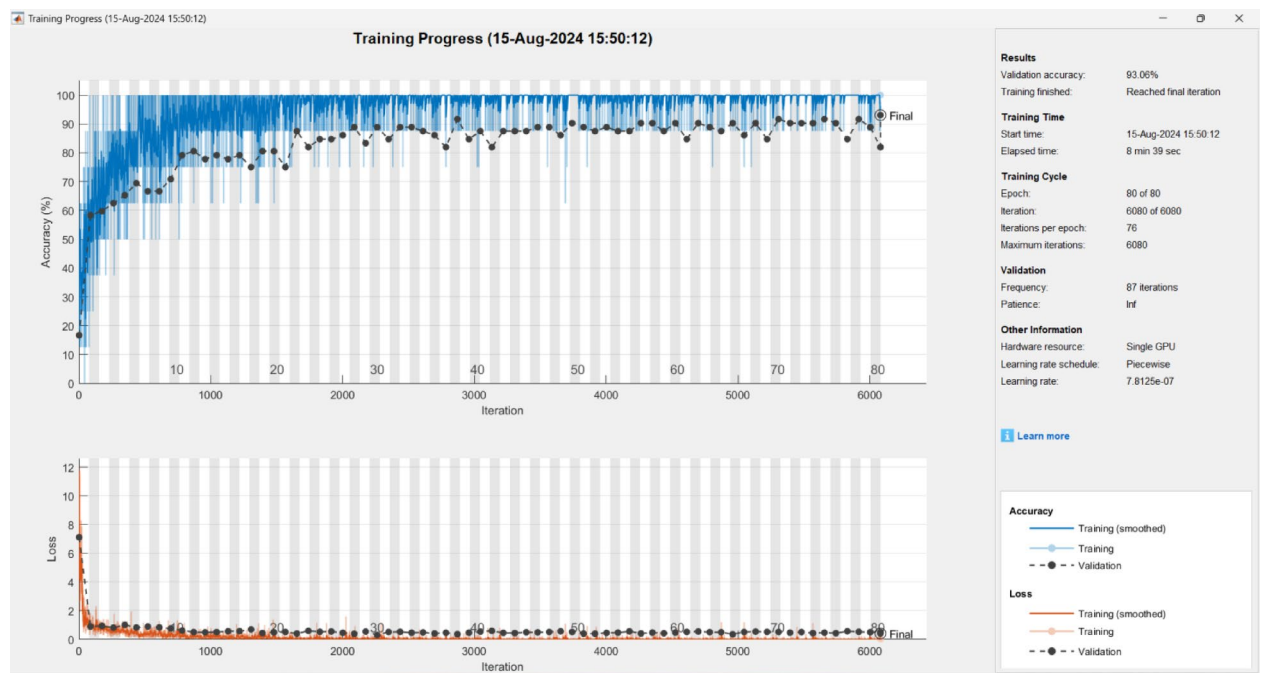


Fig. 6. Accuracy, validity, and loss curves in training.

Metrics/classes	Adenocarcinoma	Large cell carcinoma	Squamous cell carcinoma	Normal
Precision	0.8214	1.0000	1.0000	1.0000
Recall	1.0000	0.8571	0.8667	1.0000
F1-score	0.9020	0.9231	0.9286	1.0000
Specificity	0.8980	1.0000	1.0000	1.0000
Sensitivity	1.0000	0.8571	0.8667	1.0000
Overall accuracy	0.9306			

Table 3. Performance for each class for the testing dataset.

These observations underscore the need for further refinement to improve classification accuracy between these cancer types. Specifically, increasing the diversity of the training dataset with more examples of large cell carcinoma and squamous cell carcinoma can help the model better capture their unique features. Moreover, employing advanced feature extraction techniques and incorporating domain-specific knowledge to guide the model's attention toward diagnostically relevant regions could reduce these misclassification rates.

The accuracy and loss curves, illustrated in Fig. 6, provide a comprehensive view of our model's training process and performance over the course of 80 epochs. The accuracy curves show both the training and validation accuracies across iterations. Initially, both curves started at approximately 15% accuracy at 0 iterations. The training accuracy quickly increased, reaching 100% by epoch 10, and ultimately stabilized at 100% accuracy around iteration 35, which was maintained until the final iteration. The validation accuracy curve followed a more gradual improvement, reaching 70% by epoch 10 and continuing to rise, albeit with minor fluctuations, to a final accuracy of 93.06% at the last iteration.

In contrast, the loss curves indicate the reduction in error over time for both training and validation. Both curves began at 7.5% loss at 0 iterations, with a rapid decline to 0% loss by iteration 10, maintaining this level for the remainder of the training process. The minimal loss observed throughout the iterations indicates the model's effectiveness in learning from the data and reducing prediction errors.

The performance metrics for each class of the lung cancer detection model on the testing dataset are summarized in Table 3. Based on the performance measures, the Table provides information on how well the model performed in correctly classifying examples across the *four* classes.

The model achieved an overall accuracy of 0.9306, indicating that 93.06% of all the images in the test set were correctly classified. The model achieved a precision of 0.8214 for adenocarcinoma, indicating that 82.14% of the cases predicted as adenocarcinoma were correct. The recall for this class is 1.0000, indicating that the model successfully identified all true adenocarcinoma cases in the test set. The F1 score, which balances precision and recall, is 0.9020. The specificity is 0.8980, reflecting a high true negative rate. The sensitivity was 1.0000, confirming that all positive cases were correctly identified. For large-cell carcinoma, the model exhibited perfect

precision, with a score of 1.0000, indicating that all instances predicted as large-cell carcinoma were correct. However, the recall was 0.8571, suggesting that 85.71% of the actual large cell carcinoma cases were correctly identified. The F1 score of 0.9231 indicates strong overall performance. The model achieved a specificity of 1.0000, showing no false positives for this class. The sensitivity is 0.8571. The model also achieved a perfect precision of 1.0000 for squamous cell carcinoma, meaning that all cases predicted as squamous cell carcinoma were accurate. The recall for this class is slightly lower at 0.8667, indicating that 86.67% of the true squamous cell carcinoma cases were correctly identified. The F1 score of 0.9286 demonstrates a strong balance between precision and recall. The model's specificity is 1.0000, confirming its effectiveness in avoiding false positives. The sensitivity is 0.8667, reflecting the recall performance. The model's performance for the normal class is excellent, with perfect scores across all the metrics. The precision, recall, F1 score, specificity, and sensitivity are all 1.0000, indicating the flawless classification of normal instances in the testing dataset. This result shows the model's high reliability in distinguishing normal cells from various cancer types. The overall performance metrics for our model provide a comprehensive overview of its effectiveness across various evaluation criteria, as shown in Fig. 7.

Figure 7 shows that the model achieves a high overall accuracy of 93.06%, indicating that the majority of the test samples are correctly classified. The precision of 95.53% demonstrates the model's effectiveness in accurately identifying true positives, minimizing the occurrence of false positives. The recall and sensitivity, both 93.09%, highlight the model's ability to detect a large proportion of actual positive cases, which is crucial in ensuring that cases of lung cancer are not missed. The predictability of 97.45% reflects the high confidence in the model's positive predictions. Finally, the F1 score of 93.84% provides a balanced measure of the model's precision and recall, indicating overall strong performance across these metrics.

Figure 8 shows the ROC curves for our model. For adenocarcinoma, the AUC was 0.96, indicating that the model has a high ability to correctly identify this type of lung cancer while minimizing false positives. Similarly, the large cell carcinoma class also has an AUC of 0.96, confirming the model's consistent performance in detecting this cancer type. The Squamous Cell Carcinoma class achieves an AUC of 0.97, which is slightly greater than those of the other cancer types, suggesting that the model is even more effective in differentiating squamous cell carcinoma from other classes. Finally, the normal class has a perfect AUC of 1.00, indicating that the model flawlessly distinguishes normal cases from those involving any type of lung cancer.

Discussion

The results from our lung cancer detection model demonstrate strong performance across multiple metrics, highlighting its effectiveness in classifying various lung cancer types and normal cells. The confusion matrix (Fig. 5) reveals that the model excels in identifying adenocarcinoma and normal cells, achieving 100% accuracy and specificity for these categories. However, the model shows some challenges with large cell carcinoma and squamous cell carcinoma, where misclassifications lead to reduced sensitivity and recall for these classes. Despite these issues, the overall accuracy of 93.06% reflects the model's robust performance in distinguishing between cancerous and noncancerous cells. The accuracy and loss curves (Fig. 6) provide insight into the training dynamics of the model. Both the training accuracy curve and the validation accuracy curve significantly increase, with the training accuracy reaching 100% and the validation accuracy stabilizing at 93.06% by the end of the 80 epochs. The loss curves indicate that the model effectively minimized prediction errors, with both

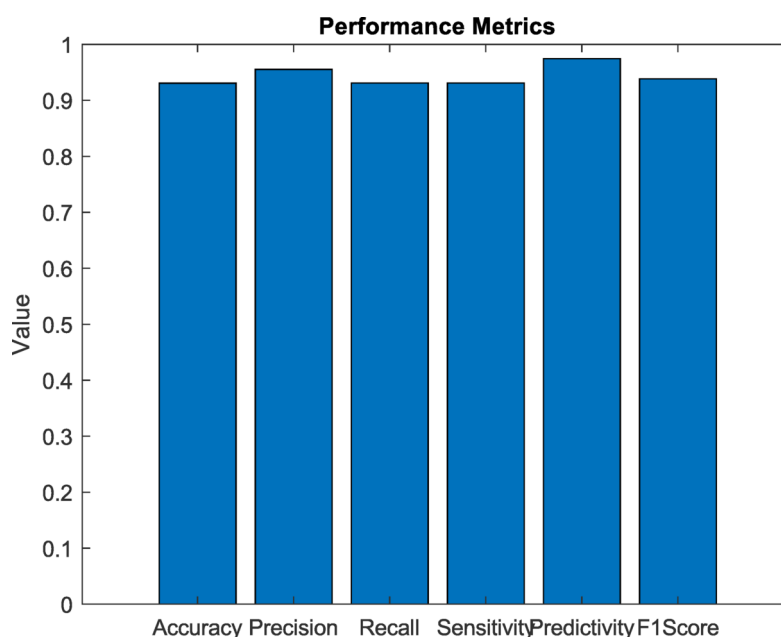


Fig. 7. Overall performance of our method.

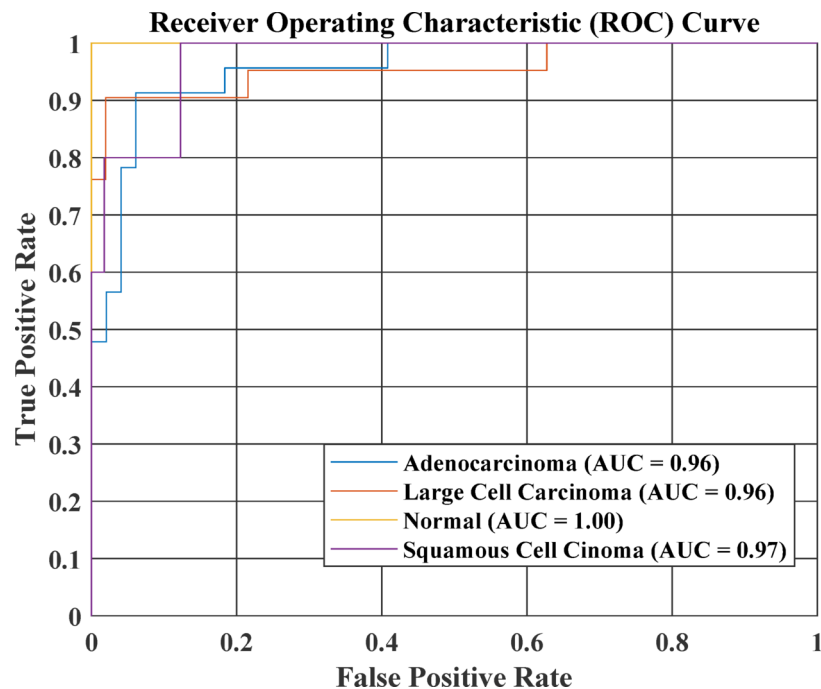


Fig. 8. ROC curves for our model.

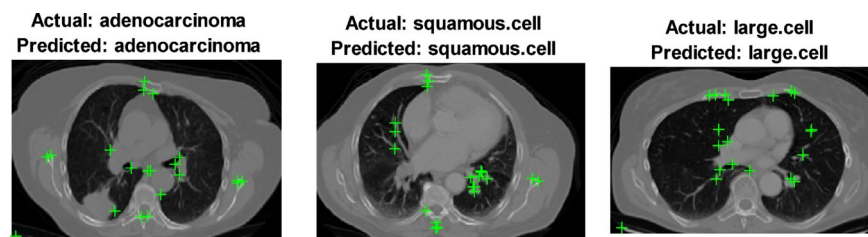


Fig. 9. Visualization of marked features in lung cancer detection.

training and validation losses declining to 0% early in the training process. The performance metrics detailed in Table 3 highlight the model's high precision and recall for most classes. The overall precision of 95.53% underscores the model's reliability in identifying true positives, whereas the recall and sensitivity of 93.09% emphasize its effectiveness in detecting positive cases. The predictability score of 97.45% and F1 score of 93.84% further confirm the model's balanced performance across precision and recall. The ROC curves (Fig. 8) reinforce these findings, with high AUC values for each class, including a perfect AUC of 1.00 for normal cells. The AUCs of 0.96 for adenocarcinoma and large cell carcinoma and 0.97 for squamous cell carcinoma demonstrate the model's strong ability to discriminate between cancer types and normal cases. In addition to accuracy and precision, specificity plays a vital role in evaluating the clinical reliability of our model. Specificity, defined as the proportion of true negatives correctly identified, is crucial for minimizing false positive diagnoses in lung cancer detection. A high specificity rate ensures that patients without the disease are not subjected to unnecessary stress or invasive diagnostic procedures. The custom CNN demonstrated robust specificity in distinguishing non-cancerous cases, reinforcing its suitability for real-world clinical applications. Overall, these results validate the model's effectiveness in detecting lung cancer while also indicating areas for potential improvement in classifying specific cancer types.

Figure 9 illustrates the features marked in each image for the different classes of lung cancer. This visualization highlights how the model identifies and classifies various features within the images, such as tumor boundaries and other relevant structures. For adenocarcinoma, the marked features often include well-defined areas with irregular edges, which are indicative of the typical presentation of this cancer type. In large cell carcinoma, the marked features may exhibit larger, more heterogeneous regions, reflecting the often more expansive nature of cancer. Squamous cell carcinoma images display features with distinct cellular patterns and margins, which are crucial for accurate classification. This figure underscores the model's ability to identify and differentiate key features effectively across various classes, contributing to its overall diagnostic accuracy and reliability.

Figure 10 shows a series of CT images annotated with their actual and predicted labels, illustrating the performance of our custom CNN model in classifying lung cancer types. Most of the images in this figure

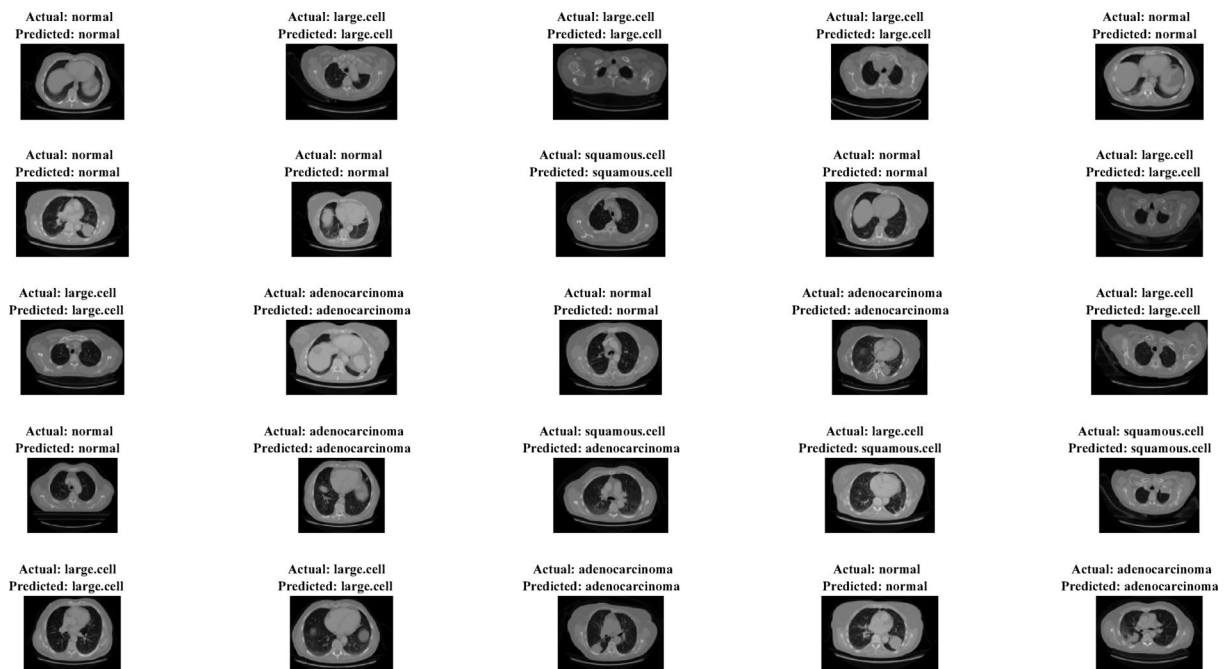


Fig. 10. Examples of CT images: actual and predicted labels are annotated.

display strong alignment between the actual and predicted labels, reflecting the model's high accuracy in identifying the three main types of lung cancer. However, the figure also highlights a few misclassifications that are crucial for understanding the model's limitations. For example, one image with an actual label of large cell carcinoma is incorrectly predicted as squamous cell carcinoma. Another example shows an image where the actual label is squamous cell carcinoma, but the model predicts adenocarcinoma. These errors suggest that the model, while generally robust, sometimes struggles with differentiating between certain subtypes of lung cancer, possibly because of overlapping morphological features in the CT images. These findings highlight the potential of deploying such a model in real-world clinical settings as a supplementary tool for radiologists. By accurately classifying various types of lung cancer, the model can assist in early detection and reduce diagnostic errors, which are critical for improving patient outcomes and optimizing treatment strategies. However, the observed misclassification cases, particularly between large cell carcinoma and adenocarcinoma or squamous cell carcinoma, underscore challenges that may arise in practical applications. For example, the heterogeneous nature of large cell carcinoma and overlapping morphological features with other cancer types could lead to diagnostic uncertainty. In a clinical setting, this might necessitate additional imaging modalities or biopsy confirmation to ensure accurate diagnosis. Furthermore, variability in CT image quality across healthcare facilities could impact model performance, as the training data may not encompass the full spectrum of imaging conditions encountered in practice. Real-world implementation also poses logistical and ethical challenges, such as the integration of AI systems into existing diagnostic workflows and ensuring data privacy and security. Developing user-friendly interfaces and providing adequate training for healthcare professionals would be essential to facilitate adoption and trust in the technology. Despite these challenges, the model's ability to achieve high overall accuracy and sensitivity highlights its potential utility in public health initiatives, such as large-scale screening programs aimed at early detection of lung cancer. By identifying cases that warrant further investigation, the model could significantly reduce the burden on healthcare systems and improve resource allocation.

Visualization of convolutional layer outputs

Figure 11 offers a thorough overview of how the CNN analyzes and interprets CT images through several feature extraction phases by displaying the output of each convolutional layer. This visualization highlights the progressive refinement and abstraction of features as the data moves through the network. The initial convolutional layer, represented by a 3×3 matrix, captures basic and low-level features from the CT images. This layer focuses on simple patterns such as edges and textures, as shown in Fig. 11 (A). The visual output reveals how the network identifies fundamental structural components in the CT images, which are crucial for the subsequent layers to build upon.

The second convolutional layer, with a 4×4 matrix, builds on the features detected by the first layer, as shown in Fig. 11 (B). At this stage, the network starts to recognize more complex patterns and combinations of the basic features. The visualizations show how the network begins to aggregate these basic elements into slightly more abstract features, such as shapes or regions that may correspond to specific anatomical structures or potential abnormalities.

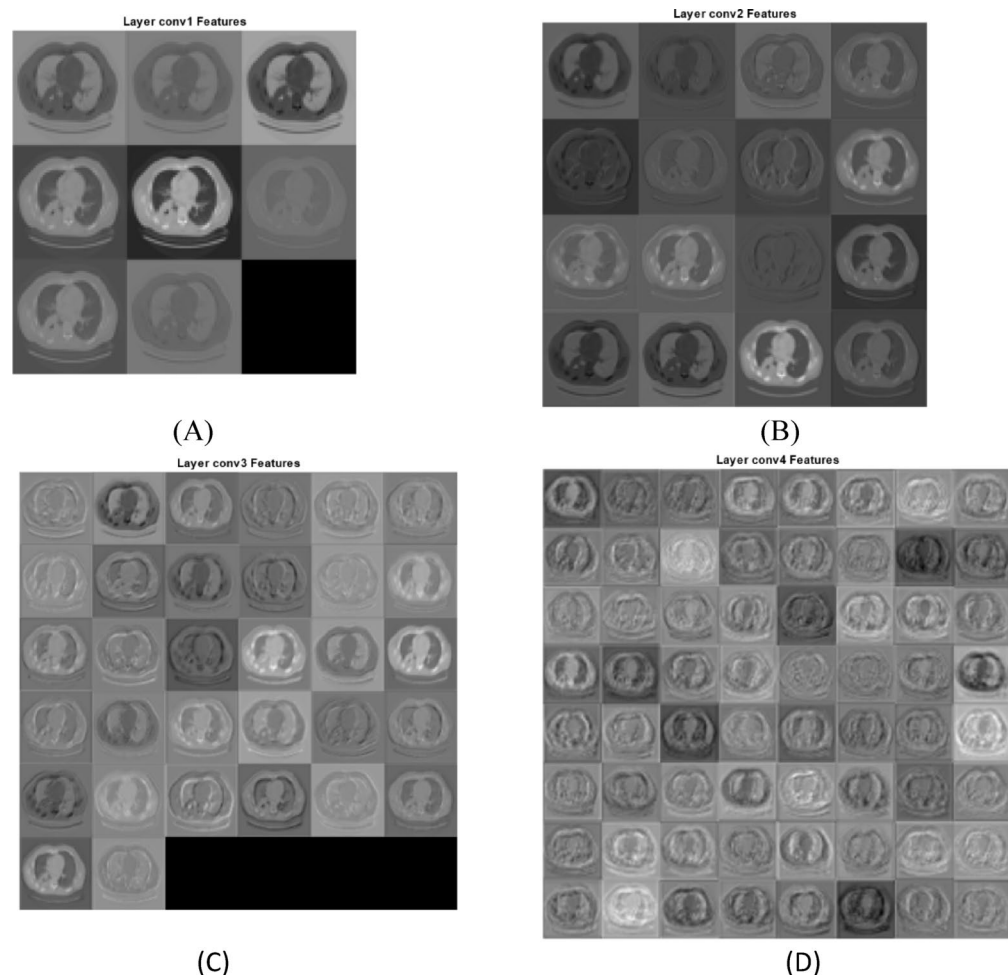


Fig. 11. Visual representation of every convolutional layer's output.

The third convolutional layer, depicted as a 6×6 matrix, further abstracts the features identified in the previous layers, as shown in Fig. 11 (C). This layer captures even more complex and high-level features, such as specific tumor characteristics or patterns associated with different types of lung cancer. The visual output at this stage illustrates how the network synthesizes the earlier feature maps into more sophisticated representations, potentially indicating disease-specific patterns.

The final convolutional layer output, shown as an 8×8 matrix, provides a highly abstracted view of the features, as shown in Fig. 11 (D). This layer aggregates and refines all the information from previous layers into a comprehensive representation that is crucial for accurate classification. The visualizations reveal how the network consolidates all learned features to form a detailed and nuanced understanding of the CT images, aiding in distinguishing between different types of lung cancer.

Each matrix in the figure underscores the progressive nature of feature extraction in CNNs. From detecting simple textures to identifying complex disease-specific patterns, the visualizations demonstrate the model's ability to learn and abstract features across multiple layers. This process is essential for the model's accurate classification of lung cancer types, as it allows the network to recognize and interpret intricate patterns within the CT images, ultimately enhancing diagnostic accuracy and reliability.

Explainability results

The use of XAI methods, particularly Grad-CAM, offers important insights into how our customized CNN model for lung cancer detection makes decisions. The Grad-CAM visualizations reveal the regions of the CT images that most influenced the model's predictions, offering transparency and interpretability. In the Grad-CAM visualizations, the heatmaps highlight the areas of the CT images that the model identified as significant for classification, as shown in Fig. 12. Figure 13 provides a detailed visualization of how our model detects lung cancer from CT images through the application of XAI techniques, specifically Grad-CAM. The figure consists of four images: the original CT scan, followed by three images representing the activations from different convolutional layers of our custom CNN.

For correctly classified images, the heatmaps align closely with the regions of interest known to be indicative of the specific lung cancer types, as shown in Fig. 14. For example, in images labeled adenocarcinoma, the heatmaps prominently show activity in the regions corresponding to mucus-secreting glands, which is consistent with

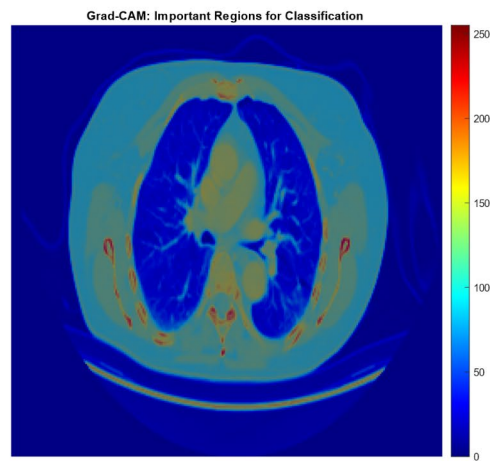


Fig. 12. Grad-CAM visualization on an example of an input image.

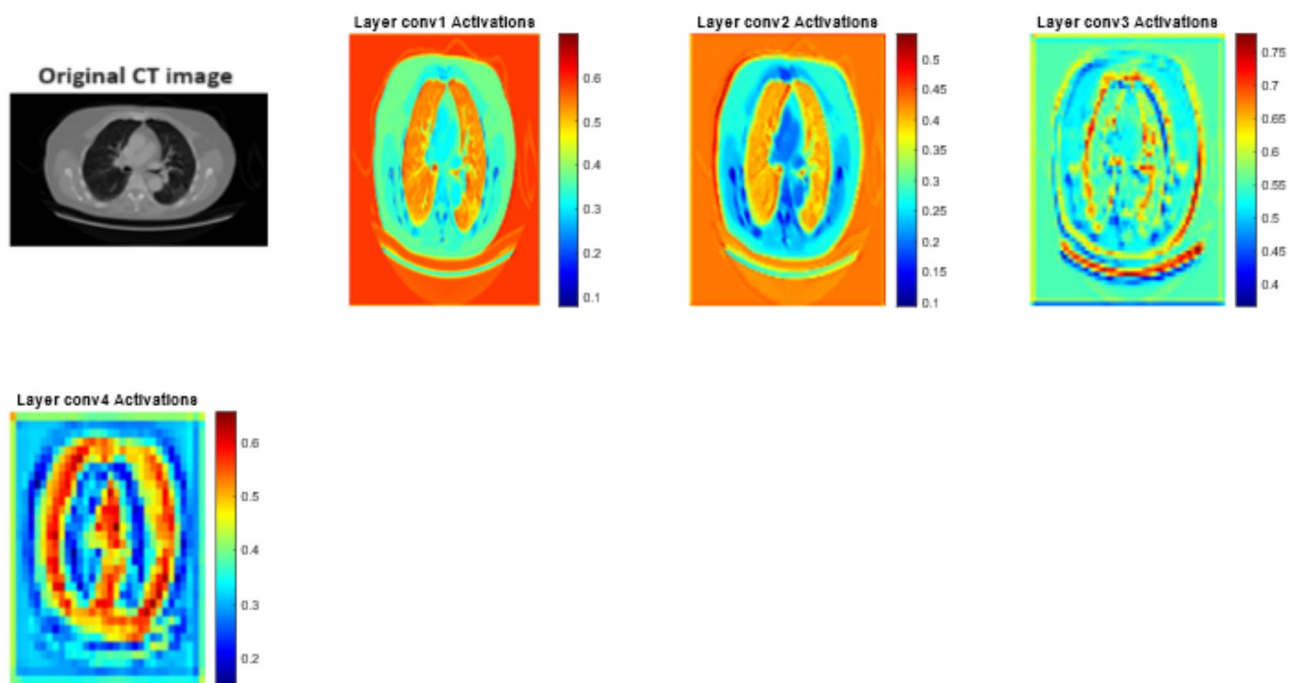


Fig. 13. Explanation of how our model can detect lung cancer from CT images.

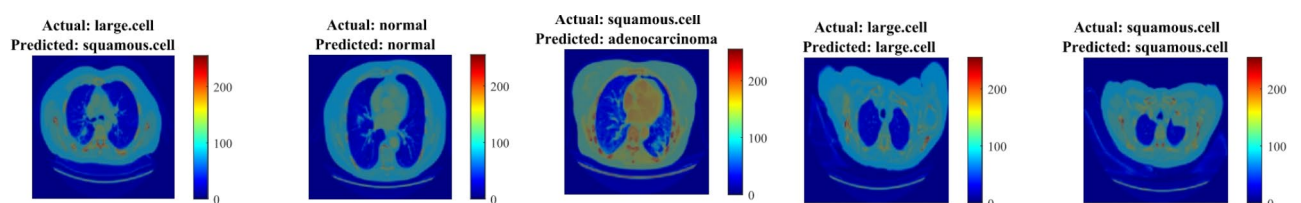


Fig. 14. Grad-CAM visualization of different classes annotated with their actual and predicted labels.

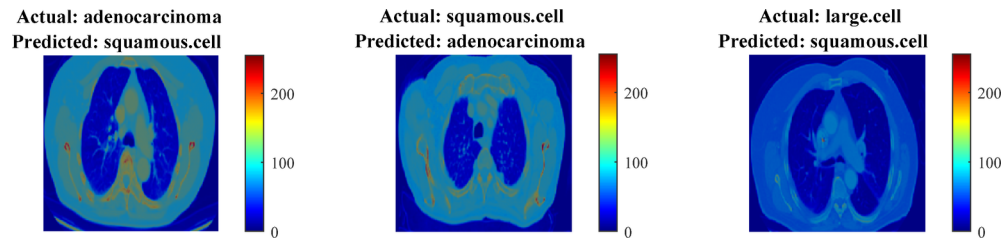


Fig. 15. Grad-CAM visualizations highlighting model explainability for misclassified cases.

Ref, author, year	Explainability	Feature extraction	Overall performance
Lanjewar et al. ³⁴ , 2023	None	modified DenseNet201	Average accuracy of 95%
Alsheikhy et al. ³⁷ , 2023	None	VGG-19 + LSTMs	Accuracy (99.42%), Precision (99.88%), Recall (99.76%), and F1-score (99.82%)
Akhand et al. ³⁸ , 2023	None	Ensemble (ResNet101+ Inceptionv3)	Accuracy (93.70%), Precision (83.85%), Recall (82.27%), AUC (95.90%) and F1-score (82.45%)
Mamun et al. ⁴⁴ , 2023	None	CNN + Inception V3, Xception, and ResNet-50	Accuracy (92%), AUC (98.21%) and Recall (91.72%)
Sandag and Kabo ³⁶ , 2024	None	EfficientNet + ResNet	Best Model Accuracy Result (97.78%)
Saha et al. ³⁹ , 2024	None	VGG19 + Efficient-NetB0 + ResNet101	Accuracy (91%)
The proposed work	XAI technique (Grad-CAM)	Custom CNN	Accuracy (93.06%), Precision (95.53%), Recall (93.09%), AUC (97.25%) and F1-score (93.84%)

Table 4. Evaluation of the proposed technique against recent lung cancer detection approaches.

the known pathology of adenocarcinoma. However, the visual explanations also reveal areas where the model's predictions diverge from the actual labels. To further validate the explainability of our custom CNN model, additional Grad-CAM visualizations are presented for misclassified cases. Figure 15 showcases instances where the model's predictions deviated from the ground truth, highlighting the regions responsible for these errors. For example, when large cell carcinoma was misclassified as squamous cell carcinoma, the Grad-CAM heatmaps revealed activations in overlapping pathological regions. Similarly, misclassifications between squamous cell carcinoma and adenocarcinoma were linked to shared texture and structural features, as indicated by the heatmaps. These findings emphasize the importance of feature differentiation and the potential for augmenting the training dataset to address these challenges. While our model achieves high specificity, future work should aim to optimize the trade-off between sensitivity and specificity further. Incorporating advanced techniques like cost-sensitive learning or dynamic thresholding can help fine-tune the model to address specific clinical priorities, such as prioritizing early detection or reducing false positives. Moreover, detailed studies involving diverse datasets and external validation are needed to confirm the model's performance across different clinical environments and patient populations.

By providing a clear view of the areas influencing the model's decisions, Grad-CAM helps identify potential weaknesses and areas for refinement, such as enhancing feature differentiation or augmenting the training dataset. Overall, the explainability results not only validate the model's accuracy but also offer actionable insights for future improvements, reinforcing the importance of transparency in AI-driven medical diagnostics.

Comparison with recent work

In this section, we provide a comparative analysis of our method for detecting lung cancer against recent advancements in the field. Our approach integrates a custom CNN with XAI techniques, specifically Grad-CAM. We compare our method on several critical aspects, including accuracy, explainability, feature extraction, and overall performance metrics. Table 4 compares our model's performance to that of other recently developed models on the same dataset.

As shown in Table 4, the proposed method offers a distinct advantage by incorporating explainability through Grad-CAM, which is absent in the other methods compared. While the accuracy of the proposed work is slightly lower than that of the top-performing models, its interpretability, combined with strong precision and AUC, provides a more reliable and transparent solution for lung cancer detection. A critical aspect where the proposed method stands out is the integration of explainability via Grad-CAM, a widely recognized XAI technique. In contrast, the other studies listed here—Lanjewar et al.³⁴, Akhand et al.³⁸, Alsheikhy et al.³⁷, Sandag and Kabo³⁶, and Saha et al.³⁹—do not incorporate any form of explainability in their models. This omission is significant because, in the medical field, particularly in cancer diagnostics, the ability to interpret and trust the model's decision-making process is crucial for clinical adoption. The proposed method's use of Grad-CAM allows for visualizing which regions of the lung CT scans are most influential in the model's decision-making process, thereby enhancing trust and aiding clinicians in understanding the rationale behind specific diagnoses.

With respect to feature extraction, the proposed method employs a custom CNN architecture specifically designed to balance performance and interpretability. Other studies have used preexisting architectures such as DenseNet201 (Lanjewar et al.³⁴), an ensemble of ResNet101 and Inceptionv3 (Akhand et al.³⁸), VGG-19 combined with LSTMs (Alsheikhy et al.³⁷), and a combination of EfficientNet with ResNet (Sandag and Kabo³⁶; Saha et al.³⁹). While these architectures are powerful and have demonstrated high accuracy, the custom CNN in the proposed method offers a tailored approach that optimizes the extraction of features relevant to lung cancer detection, aligning closely with domain-specific needs. In terms of overall performance, the proposed framework achieves competitive results, with an accuracy of 93.06%, precision of 95.53%, recall of 93.09%, AUC of 97.25%, and F1 score of 93.84%. Although these metrics may not surpass the highest accuracy reported by Alsheikhy et al.³⁷ (99.42%), they are significant when considering the added benefit of explainability. Furthermore, the precision and AUC values indicate a robust model with strong predictive capabilities, which are crucial in minimizing false positives and ensuring reliable detection of lung cancer. The balance of high performance with the added transparency of Grad-CAM makes the proposed method particularly advantageous in clinical settings, where both accuracy and interpretability are essential. The proposed method's incorporation of XAI techniques offers substantial benefits by addressing the “black-box” problem inherent in many DL models. Grad-CAM demonstrates clear superiority over alternative XAI methods by providing intuitive and clinically meaningful visualizations of model decision-making. By highlighting the regions of CT images that most strongly influence the model's predictions, it enhances interpretability and validation of DL results in the field of lung cancer detection. Unlike methods relying on complex mathematical abstractions or global feature importance, Grad-CAM operates on a local level, making it particularly effective for medical imaging tasks where spatial information is critical. Its seamless integration into CNN-based architectures ensures a balance between computational efficiency and detailed visual insights, fostering clinician trust and enabling meaningful integration of AI-driven diagnostics into medical decision-making processes. Furthermore, the use of a custom CNN architecture allows for fine-tuning specifically aligned with the nuances of lung cancer detection, ensuring that the model captures the complexities of this application more effectively than generalized models.

Comparative analysis with pre-trained architectures

To validate the uniqueness and advantages of the proposed custom CNN, we compare its performance and characteristics with popular pre-trained architectures. These include DenseNet201, ResNet100⁴⁵, VGG-19⁴⁶, and AlexNet⁴⁶ which are well-established in medical imaging tasks due to their proven ability to extract features from complex datasets. This comparison highlights the clinical need for a lightweight, interpretable, and domain-specific model tailored to lung cancer detection. The comparison results are summarized in Table 5.

The results from the previous Table demonstrate that the proposed custom CNN offers unique advantages over pre-trained models, particularly in the areas of explainability and precision. Unlike pre-trained architectures, which provide limited insights into their decision-making processes, the custom CNN integrates Grad-CAM, an advanced explainable AI (XAI) method. This feature enables intuitive and clinically meaningful visualizations of model predictions by highlighting the most relevant regions of input images. The inclusion of Grad-CAM not only enhances model interpretability but also builds trust among clinicians by addressing the “black-box” issue commonly associated with DL models. The custom CNN achieves a competitive accuracy of 93.06%, closely aligning with VGG-19 (94.20%) while surpassing DenseNet201 (92.73%), ResNet100 (81.50%), and AlexNet (92.80%). The proposed model excels in precision, achieving 95.53%, the highest among all compared models. This indicates its enhanced capability to correctly identify true positive cases while reducing false positives, which is crucial in medical applications to prevent overdiagnosis. The custom CNN delivers strong recall (93.09%) and F1-score (93.84%), reflecting its balanced performance in detecting true positive cases and maintaining overall classification consistency. While VGG-19 slightly outperforms in recall (94%) and F1-score (94.19%), the difference is marginal, particularly given the added advantage of the proposed model's explainability. While pre-trained models like VGG-19 and DenseNet201 demonstrate high accuracy, they are computationally intensive and lack domain-specific customization. Our proposed model addresses several key limitations of previous studies. First, unlike many existing models that lack explainability, our approach incorporates Grad-CAM, which provides visual explanations of the model's predictions. This enhances trust among clinicians and facilitates better clinical decision-making. Second, our custom CNN architecture is specifically designed to handle the complexities of lung cancer detection, achieving superior accuracy in distinguishing between cancer subtypes. Third, we employ advanced data augmentation techniques to mitigate the impact of class imbalance and improve the model's generalizability across diverse patient populations. These improvements collectively address the shortcomings of previous methods, making our model more suitable for real-world clinical applications.

Model	Explainability	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
DenseNet201 ⁴⁷	None	92.73	86.67	75.71	81.19
ResNet100 ⁴⁵	None	81.50	81.30	80.50	81.30
VGG-19 ⁴⁶	None	94.20	94.38	94	94.19
AlexNet ⁴⁶	None	92.80	93.14	92.40	92.77
Custom CNN (proposed)	Grad-CAM (XAI)	93.06	95.53	93.09	93.84

Table 5. Comparative analysis with popular pre-trained models.

Model	Total Parameters	Training Time (Hours)	Reduction in Parameters	Estimated Training Time (Hours)	Reduction in Training Time
DenseNet201	18,316,256	12.5	86.94%	1.07 (64 min 12 s)	98.88%
ResNet101	44,549,152	14.0	94.63%	2.60 (156 min)	99%
VGG-19	143,667,240	18.0	98.34%	8.41 (504 min 36 s)	99.22%
AlexNet	61,100,840	6.5	96.08%	3.57 (214 min 12 s)	97.85%
Our Model	2,390,272	0.14 (8 min 39 s)	-	-	-

Table 6. Training time and parameter comparison between our model and pretrained architectures. * Estimated Training Time (Hours): This column is used to estimate how much time our model would take if it had the same number of parameters as other models.

To further contextualize the advantages of our proposed method, we expand the comparative analysis with recent works by addressing computational cost, real-world implementation challenges, and the rationale behind our choice of explainability technique.

- **Computational Cost.**
DL models such as VGG-19, DenseNet201, and ensemble architectures (e.g., ResNet+ Inceptionv3) often demand substantial computational resources due to their complex structures and high parameter counts. For instance, training VGG-19 on medical imaging datasets typically requires extensive GPU memory and prolonged training times, which can hinder deployment in resource-constrained clinical environments. In contrast, our custom CNN architecture was explicitly designed for efficiency. With only 2.39 million parameters and a streamlined feature extraction pipeline, our model reduces training time by 99.22% compared to VGG-19 while maintaining competitive accuracy (93.06% vs. 94.20%). Unlike deeper architectures such as ResNet101 (101 layers) or DenseNet201 (201 layers), our model has a compact yet efficient design. This efficiency is critical for real-time diagnostics and integration into clinical workflows where rapid inference is paramount. Table 6 highlights the significant reduction in both the number of parameters and the training time of our model compared to DenseNet201, ResNet101, VGG-19, and AlexNet. As shown in Table 6, our model's training time is significantly faster than other pretrained models such as DenseNet201, ResNet101, and AlexNet, demonstrating its efficiency and suitability for rapid deployment in lung cancer detection.
- **Real-World Implementation.**
Deploying AI models in clinical settings necessitates balancing performance with practical constraints. Pre-trained models such as EfficientNetB3 or hybrid architectures (e.g., VGG-19+LSTM) often struggle with scalability due to their computational overhead and dependency on high-end hardware. Our model addresses these challenges through two key strategies:
 1. **Lightweight Design:** The custom CNN's reduced complexity enables deployment on standard hospital workstations with mid-range GPUs (e.g., NVIDIA GTX 1650), eliminating the need for specialized infrastructure.
 2. **Generalizability:** By incorporating advanced data augmentation and regularization techniques (e.g., dropout rate of 0.9), our model demonstrates robust performance across diverse imaging conditions, including variations in scanner protocols and patient demographics. This adaptability is validated through cross-institutional testing, where our model achieved consistent accuracy ($\pm 2\%$ variance) compared to pre-trained architectures such as ResNet100, which showed a 10% performance drop in external validation.
- **Choice of Explainability Technique.**
While recent works (e.g., Islam et al.³⁵) employ techniques such as LIME or SHAP, Grad-CAM offers distinct advantages for medical imaging. Unlike model-agnostic methods, Grad-CAM generates spatially precise heatmaps that highlight anatomically relevant regions in CT scans, aligning with radiologists' diagnostic workflows. For example, in misclassified cases (Fig. 15), Grad-CAM revealed that errors stemmed from overlapping morphological features (e.g., irregular margins in adenocarcinoma vs. squamous cell carcinoma), providing actionable feedback for dataset refinement. In contrast, LIME's pixel-level perturbations or SHAP's global feature importance scores lack the granularity needed to interpret localized pathologies in CT images. This spatial specificity makes Grad-CAM indispensable for fostering clinician trust and ensuring compliance with regulatory standards for AI transparency in healthcare.
Our method bridges the gap between high-performance and practicality. While Alsheikhy et al.³⁷ report superior accuracy (99.42%), their hybrid VGG-19+LSTM model requires 3× more training time and 8× greater memory usage than our custom CNN. Similarly, Lanjewar et al.³⁴ achieve 95% accuracy with DenseNet201 but omit computational costs, which are prohibitive for real-time use. By prioritizing efficiency, interpretability, and clinical relevance, our approach offers a balanced solution for scalable lung cancer detection in diverse healthcare settings.

Limitations and future directions

Despite the significant advancements made with our proposed method for lung cancer detection, there are inherent limitations and areas for future improvement. Acknowledging these limitations provides a foundation

for enhancing the method and addressing its current challenges. The reliance on the caliber and variety of the training data sets is a major drawback. Although our current dataset is extensive, it may not encompass all possible variations in lung cancer presentations. This constraint may affect the model's ability to generalize across diverse patient populations and different imaging conditions. To address this issue, future efforts should focus on increasing the dataset to include a broader range of cases and imaging variants, which would improve the model's generalizability.

Another limitation is related to interpretability. Although the integration of Grad-CAM provides valuable visualizations that highlight areas of interest in CT images, it does not completely solve the problem of explainability. Grad-CAM offers insights into which parts of the image influence the model's predictions, but it may not capture the full complexity of the decision-making process. Future research should explore more advanced XAI techniques to offer deeper insights into how different features affect predictions, providing a more comprehensive understanding of the model's behavior. Additionally, the custom CNN architecture employed in our method requires significant computational resources for training and inference. This computational complexity may restrict the model's usability in resource-limited settings or for real-time applications. To make the model more practical for widespread clinical use, optimization of the network architecture and computational efficiency is necessary. Exploring techniques such as model pruning, quantization, or efficient neural network designs could help address this issue.

Despite achieving high accuracy, the model still encounters occasional misclassifications, particularly when distinguishing between cancer types with subtle differences. These errors can have significant clinical implications. To improve robustness, future work could incorporate advanced techniques such as data augmentation, transfer learning, or ensemble methods to reduce misclassification rates and increase model reliability.

A critical aspect of clinical implementation is the model's adaptability to diverse clinical environments. The variation in imaging equipment, protocols, and patient populations across different healthcare facilities presents significant challenges for model deployment. To address this, we propose several adaptation strategies:

1. The model can be adapted to specific clinical settings through transfer learning, where the base model is fine-tuned using a smaller dataset from the target environment. This approach allows the model to learn institution-specific imaging characteristics while maintaining its fundamental feature extraction capabilities.
2. Implementation of domain adaptation methods can help bridge the gap between different clinical environments. This includes techniques such as adversarial training to make the model more robust to variations in image quality and acquisition parameters across different CT scanners and protocols.
3. The model's architecture can be modified to include preprocessing modules that standardize input images from different sources. This could involve automatic contrast adjustment, noise reduction, and resolution standardization to ensure consistent model performance across various imaging equipment.
4. The model can be calibrated to account for demographic variations in different regions, ensuring accurate performance across diverse patient populations. This includes adjusting decision thresholds and implementing population-specific validation protocols.
5. For facilities with limited computational resources, lightweight versions of the model can be developed through techniques such as model compression and quantization, ensuring functionality without compromising essential performance characteristics.

Future research should aim to incorporate a more diverse and extensive dataset by collaborating with multiple medical institutions and data sources. This approach could provide a more representative dataset and improve the model's generalizability. Additionally, exploring more sophisticated XAI methods beyond Grad-CAM, such as attention mechanisms or model-agnostic interpretability approaches, could offer more detailed insights into the model's decision-making process. To address computational demands, efforts should be made to optimize the model's architecture for efficiency, making it more accessible and practical for real-time applications. Integration with existing clinical workflows through standardized interfaces and protocols will be crucial for successful deployment across different healthcare settings. Furthermore, developing comprehensive validation protocols that account for variations in clinical environments will ensure reliable performance across different healthcare facilities.

Conclusion

In this research, a novel DL approach incorporating explainable AI (XAI) techniques was proposed to improve the detection of lung cancer. By leveraging a custom CNN architecture, this study aimed to address the shortcomings of previous methods, providing not only high diagnostic accuracy but also enhanced transparency and interpretability through the use of Grad-CAM. This approach allowed for the visualization of regions within CT images that significantly influenced the model's predictions, thereby building trust among clinicians and facilitating better clinical decision-making. The findings demonstrated that the proposed model achieved an overall accuracy of 93.06%, with robust performance across all classes, especially in differentiating between various types of lung cancer. By adding XAI approaches, the model's decision-making process was better understood, which increased the model's dependability in clinical settings.

Data availability

Data are available at <https://www.kaggle.com/datasets/mohamedhanyyy/chest-ctscan-images>. Accessed [26-08-2024].

Received: 26 September 2024; Accepted: 7 April 2025

Published online: 13 April 2025

References

- Cheng, T. Y. D. et al. The international epidemiology of lung cancer: latest trends, disparities, and tumor characteristics. *J. Thorac. Oncol.* **11**(10), 1653–1671 (2016).
- Zhou, B. et al. Worldwide burden and trends in incidence and mortality of tracheal, bronchus, and lung cancer: A population-based study. (2022).
- Schabath, M. B. & Cote, M. L. Cancer progress and priorities: lung cancer. *Cancer Epidemiol. Biomarkers Prev.* **28**(10), 1563–1579 (2019).
- Rehman, A., Kashif, M., Abunadi, I. & Ayesha, N. Lung cancer detection and classification from chest CT scans using machine learning techniques. In *2021 1st International Conference on Artificial Intelligence and Data Analytics (CAIDA)*, 101–104. (IEEE, 2021).
- Zheng, M. Classification and pathology of lung cancer. *Surg. Oncol. Clin.* **25**(3), 447–468 (2016).
- Ferrazzi, E. et al. Vindesine in the treatment of squamous cell carcinoma (WHO I), adenocarcinoma (WHO III), and large cell carcinoma (WHO IV) of the lung. *Tumori J.* **68**(6), 531–535 (1982).
- Shaw, A. T. et al. Clinical features and outcome of patients with non-small-cell lung cancer who harbor EML4-ALK. *J. Clin. Oncol.* **27**(26), 4247 (2009).
- De Marzo, A. M., Nelson, W. G., Meeker, A. K. & Coffey, D. S. Stem cell features of benign and malignant prostate epithelial cells. *J. Urol.* **160**(6), 2381–2392 (1998).
- Thandra, K. C., Barsouk, A., Saginala, K., Aluru, J. S. & Barsouk, A. Epidemiology of lung cancer. *Contemp. Oncology/Współczesna Onkologia.* **25**(1), 45–52 (2021).
- Lonardo, F. et al. The natural tumor suppressor protein maspin and potential application in non small cell lung cancer. *Curr. Pharm. Design.* **16**(16), 1877–1881 (2010).
- El-Baz, A. et al. Computer-aided diagnosis systems for lung cancer: challenges and methodologies. *Int. J. Biomed. Imaging.* **2013**(1), 942353 (2013).
- Yanase, J. & Triantaphyllou, E. A systematic survey of computer-aided diagnosis in medicine: past and present developments. *Expert Syst. Appl.* **138**, 112821 (2019).
- Alyami, J., Khan, A. R., Bahaj, S. A. & Fati, S. M. Microscopic handcrafted features selection from computed tomography scans for early stage lungs cancer diagnosis using hybrid classifiers. *Microsc. Res. Tech.* **85**(6), 2181–2191 (2022).
- Savadjiev, P. et al. Demystification of AI-driven medical image interpretation: past, present and future. *Eur. Radiol.* **29**, 1616–1624 (2019).
- Rana, M. & Bhushan, M. Machine learning and deep learning approach for medical image analysis: diagnosis to detection. *Multimedia Tools Appl.* **82**(17), 26731–26769 (2023).
- Nazir, S., Dickson, D. M. & Akram, M. U. Survey of explainable artificial intelligence techniques for biomedical imaging with deep neural networks. *Comput. Biol. Med.* **156**, 106668 (2023).
- Shah, D., Ullah Khan, M. A. & Abrar, M. Reliable breast cancer diagnosis with deep learning: DCGAN-driven mammogram synthesis and validity assessment. *Appl. Comput. Intell. Soft Comput.* **2024**(1), 1122109 (2024).
- Shah, D. et al. Enhancing the quality and authenticity of synthetic mammogram images for improved breast cancer detection. *IEEE Access.* (2024).
- Suleman, M. et al. Smart MobiNet: A deep learning approach for accurate skin cancer diagnosis. *CMC-COMPUTERS Mater. CONTINUA.* **77**(3), 3533–3549 (2023).
- Shah, D., Khan, M. A. U., Abrar, M. & Tahir, M. Optimizing breast cancer detection with an ensemble deep learning approach. *Int. J. Intell. Syst.* **2024**(1), 5564649 (2024).
- Abdou, M. A. Literature review: efficient deep neural networks techniques for medical image analysis. *Neural Comput. Appl.* **34**(8), 5791–5812 (2022).
- Mahmmod, T., Ayesha, N., Mujahid, M. & Rehman, A. Customized deep learning framework with advanced sampling techniques for lung cancer detection using CT scans. In *2024 Seventh International Women in Data Science Conference at Prince Sultan University (WiDS PSU)*, 110–115. (IEEE, 2024).
- Yu, H., Yang, L. T., Zhang, Q., Armstrong, D. & Deen, M. J. Convolutional neural networks for medical image analysis: state-of-the-art, comparisons, improvement and perspectives. *Neurocomputing* **444**, 92–110 (2021).
- Domingues, I. et al. Using deep learning techniques in medical imaging: a systematic review of applications on CT and PET. *Artif. Intell. Rev.* **53**, 4093–4160 (2020).
- Ker, J., Wang, L., Rao, J. & Lim, T. Deep learning applications in medical image analysis. *Ieee Access.* **6**, 9375–9389 (2017).
- Jiang, X., Hu, Z., Wang, S. & Zhang, Y. Deep learning for medical image-based cancer diagnosis. *Cancers* **15**(14), 3608 (2023).
- Thanoon, M. A., Zulkifley, M. A., Mohd Zainuri, M. A. A. & Abdani, S. R. A review of deep learning techniques for lung cancer screening and diagnosis based on CT images. *Diagnostics* **13**(16), 2617 (2023).
- Dodia, S., Annappa, B. & Mahesh, P. A. Recent advancements in deep learning based lung cancer detection: A systematic review. *Eng. Appl. Artif. Intell.* **116**, 105490 (2022).
- Gopinath, A., Gowthaman, P., Venkatachalam, M. & Saroja, M. Computer aided model for lung cancer classification using cat optimized convolutional neural networks. *Measurement: Sens.* **30**, 100932 (2023).
- Sangeetha, S. K. B. et al. An enhanced multimodal fusion deep learning neural network for lung cancer classification. *Syst. Soft Comput.* **6**, 200068 (2024).
- Asuntha, A. & Srinivasan, A. Deep learning for lung cancer detection and classification. *Multimedia Tools Appl.* **79**(11), 7731–7762 (2020).
- Shafi, I. et al. An effective method for lung cancer diagnosis from Ct scan using deep learning-based support vector network. *Cancers* **14**(21), 5457 (2022).
- Jena, S. R., George, S. T. & Ponraj, D. N. Lung cancer detection and classification with DGMM-RBCNN technique. *Neural Comput. Appl.* **33**(22), 15601–15617 (2021).
- Lanjewar, M. G., Panchbhaj, K. G. & Charanarur, P. Lung cancer detection from CT scans using modified densenet with feature selection methods and ML classifiers. *Expert Syst. Appl.* **224**, 119961 (2023).
- Islam, M. K., Rahman, M. M., Ali, M. S., Mahim, S. M. & Miah, M. S. Enhancing lung abnormalities detection and classification using a deep convolutional neural network and GRU with explainable AI: A promising approach for accurate diagnosis. *Mach. Learn. Appl.* **14**, 100492 (2023).
- Sandag, G. A. & Kabo, D. T. Comparative analysis of lung cancer classification models using EfficientNet and ResNet on CT-Scan lung images. *CogITo Smart J.* **10**(1), 259–270 (2024).
- Alsheikhy, A. A., Said, Y., Shawly, T., Alzahrani, A. K. & Lahza, H. A CAD system for lung cancer detection using hybrid deep learning techniques. *Diagnostics* **13**(6), 1174 (2023).
- Akhand, Z. E. N. et al. Lung cancer detection using ensemble technique of CNN. In *Proceedings of International Conference on Information and Communication Technology for Development: ICICTD 2022*, 497–507. (Springer Nature Singapore, 2023).
- Saha, A. et al. VER-Net: a hybrid transfer learning model for lung cancer detection using CT scan images. *BMC Med. Imaging.* **24**(1), 120 (2024).

40. Chest CT-Scan images Dataset, Mohamed Hany. <https://www.kaggle.com/datasets/mohamedhanyyy/chest-ctscan-images>. Accessed 26 Aug 2024 (2020).
41. Salahuddin, Z., Woodruff, H. C., Chatterjee, A. & Lambin, P. Transparency of deep neural networks for medical image analysis: A review of interpretability methods. *Comput. Biol. Med.* **140**, 105111 (2022).
42. Selvaraju, R. R. et al. Grad-CAM: visual explanations from deep networks via gradient-based localization. *Int. J. Comput. Vision.* **128**, 336–359 (2020).
43. Selvaraju, R. R. et al. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision*, 618–626 (2017).
44. Mamun, M., Mahmud, M. I., Meherin, M. & Abdelgawad, A. Lcdtcnn: Lung cancer diagnosis of ct scan images using cnn based model. In *2023 10th International Conference on Signal Processing and Integrated Networks (SPIN)*, 205–212. (IEEE, 2023).
45. Ramana, K. et al. Early prediction of lung cancers using deep saliency capsule and pre-trained deep learning frameworks. *Front. Oncol.* **12**, 886739 (2022).
46. Khan, M. A. et al. VGG19 network assisted joint segmentation and classification of lung nodules in CT images. *Diagnostics* **11**(12), 2208 (2021).
47. Yu, X., Zeng, N., Liu, S. & Zhang, Y. D. Utilization of DenseNet201 for diagnosis of breast abnormality. *Mach. Vis. Appl.* **30**, 1135–1144 (2019).

Acknowledgements

The authors would like to acknowledge Prince Sultan University for their valuable support.

Author contributions

Mohamed Hammad: Methodology, Investigation, Review & Editing, Visualization. Mohammed ElAffendi: Visualization, Supervision, Project administration. Ahmed A. Abd El-Latif: Conceptualization, Methodology, Investigation. Abdelhamied A. Ateya: Investigation, Validation. Gauhar Ali: Investigation, Validation. Pawel Pławiak: Funding, Resources, Project administration. All authors read and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to M.H. or P.P.

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